

Julius Sanchez

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EDUCATION

University of Washington — B.S. in Computer Science

Expected Spring 2028

GPA: 3.8/4.0

EXPERIENCE

Software Engineering Intern, MeowTalk — Seattle, WA

Summer 2023 – Winter 2024

- Built a **full-stack audio annotation platform** (Python, FastAPI, React) for structured, multi-label ground-truth labeling of machine learning training data
- Used the platform to label **20,000+ audio files** across 20 categories and 4 acoustic environments
- Improved audio classification model accuracy by **9%** across 20 labels and 5 classification domains by delivering clean, structured ground-truth datasets

PROJECTS

Pyxis — Python, UMAP, Leiden, WebGL (regl) · github.com/Juliuscease123221/Pyxis

- Mapped **56,129 Steam games** by semantic similarity using hybrid text + community-tag embeddings and UMAP; on a held-out test, the hybrid beat text-only on tag prediction (56.0% vs 50.1%)
- Built a **1,117-node topic hierarchy** with recursive Leiden clustering and sibling-scoped c-TF-IDF labels, reaching exact cluster nesting (1.000 vs 0.90 for multi-resolution Leiden)
- Engineered a binary tile engine (**1M points in 1.07 s**) and a single-draw-call WebGL renderer with level-of-detail labels; cut first-load payload **18.5×** to 195 KB (first frame in 1.83 s at 5 Mbps)

Iroshine — Python, Manim, NumPy · github.com/Juliuscease123221/Iroshine

- Built a Python library that turns unmodified LeetCode solutions into **animated, sound-synced videos** by rewriting function source to log every list operation, then replaying the event log as Manim animations
- Published the rendered algorithm videos on social media, reaching **50K+ impressions**

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: FastAPI, React, NumPy, pandas, Manim, UMAP, WebGL (regl)

Tools: Git, Docker, AWS, Linux

ACTIVITIES

Competitive Programming Club (ICPC) — University of Washington

Sep 2025 – Present

- Solve algorithmic problems in Python with a club that competes in the ICPC

Robotics Team Co-Captain

2023 – 2025

- Co-led the team to a **regional championship win** and a **World Championship qualification**
- Ran match strategy with one other teammate: scouted opponent and alliance robots and planned strategies around their strengths and weaknesses